

Necessity Entrepreneurship and Misallocation

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Motivation

- ▶ Aggregate entrepreneurship is often higher in lower-income economies.
- ▶ But composition matters: necessity entrepreneurship is much more prevalent where formal jobs and insurance are limited.
- ▶ Our goal is to separate *opportunity* vs *necessity* entrepreneurship using optimal behavior, not self-reported motives.
- ▶ Central policy question: how unemployment insurance (UI) affects occupational sorting and misallocation.

Definitions and Identification

- ▶ Existing empirical proxy: previous employment status (employed = opportunity, unemployed = necessity).
- ▶ Limitation: this can misclassify agents with strategic entry timing.
- ▶ In our model, necessity entrepreneurship is defined by a **counterfactual occupational choice**:
 - ▶ Necessity entrepreneur: chooses entrepreneurship only when unemployed.
 - ▶ Opportunity entrepreneur: chooses entrepreneurship regardless of employment status.
- ▶ This maps naturally to cutoff rules by education and labor-market status.

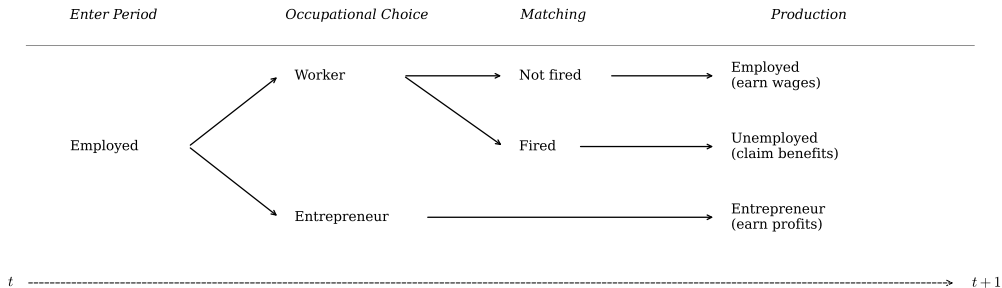
Research Questions

- ▶ How much of entrepreneurship is necessity-driven in equilibrium?
- ▶ Does stronger UI reduce necessity entry and improve allocation?
- ▶ Are effects concentrated by education group?
- ▶ Do financing and matching frictions amplify these effects?

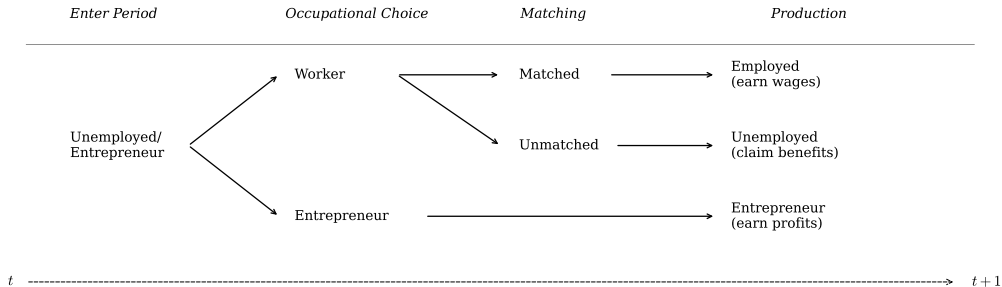
Economic Environment

- ▶ Infinite-horizon economy with occupational choice and search frictions.
- ▶ State variables: (h, a, x, z, i_w) .
- ▶ Education groups: $h \in \{P, S, T\}$.
- ▶ Labor status: unemployed, employed, entrepreneur.
- ▶ Occupational choice each period: worker vs entrepreneur.

Timing: Previously Employed



Timing: Previously Unemployed / Entrepreneur



Firm Problem

$$\max_n xk^\alpha(zn)^\gamma - zw(1 + \psi\tau)n - rk - \kappa \frac{\rho zn}{q(\theta)}$$

- ▶ Output: span-of-control technology with managerial talent x .
- ▶ Hiring/search costs rise when vacancy filling is harder ($q(\theta)$ lower).
- ▶ Main implication: weaker labor markets raise effective labor costs and shrink efficient firm scale.

Household Problem and Worker Income Risk

$$V(a, h, z, x, i_w) = \max_{\{a', c, \mathcal{I}_e\}} \{\dots\}$$

$$\tilde{w}_n = (1 - \rho)\{zw(1 - (1 - \psi)\tau)\} + \rho b, \quad \tilde{w}_u = p(\theta)\{zw(1 - (1 - \psi)\tau)\} + (1 - p(\theta))b$$

- ▶ Employed workers face separation risk ρ .
- ▶ Unemployed/ex-entrepreneurs face job-finding risk $p(\theta)$.
- ▶ Different continuation risks generate necessity vs opportunity margins.

Cutoff Characterization

- ▶ Let $\hat{h}_{i_w=1}$ and $\hat{h}_{i_w \neq 1}$ be occupation cutoffs.
- ▶ Necessity region: $\hat{h}_{i_w \neq 1} < h < \hat{h}_{i_w=1}$.
- ▶ Opportunity region: $h \geq \hat{h}_{i_w=1}$.
- ▶ Six cutoffs in total: two per education group (P, S, T) .

Calibration: Key Parameters

Parameter	Value	Interpretation
β	0.94	Discount factor
σ	2.0	Risk aversion
α	0.3207	Capital share
γ	0.4693	Labor share
ρ	$\{0.0383, 0.0288, 0.0136\}$	Separation by education
z	$\{1.0, 1.31, 1.61\}$	Wage premium by education
$\bar{m}$	0.8	Matching efficiency
κ	$[0.03, 0.045] \cdot w$	Search cost scale
b	$0.40 \cdot w$	UI replacement benchmark

Baseline Equilibrium (UI=0.4)

Outcome	Model	Notes
Entrepreneurship (avg)	0.145	baseline
Entrepreneurship (low/mid/high)	0.143 / 0.137 / 0.161	heterogeneity by education
Firm size (avg)	21.877	baseline
Firm size (low/mid/high)	9.682 / 18.007 / 33.417	steep by education

Counterfactual 1: No Unemployment Risk

Outcome	Baseline	No unemployment
Entrepreneurship (avg)	0.145	0.197
Entrepreneurship (low)	0.143	0.207
Entrepreneurship (mid)	0.137	0.184
Entrepreneurship (high)	0.161	0.216
Firm size (avg)	21.877	24.126

Interpretation: labor-market risk is a key driver of necessity entrepreneurship.

Counterfactual 2: UI Level Changes

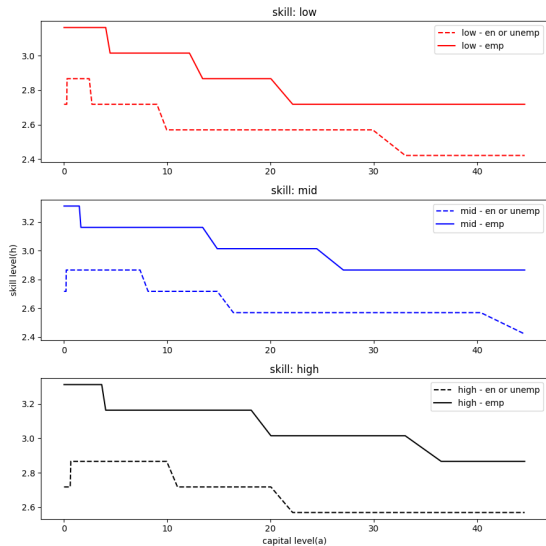
Outcome	UI=0.4	UI=0.3	UI=0.0	UI=0.6
Entrepreneurship (avg)	0.145	0.145	0.193	0.144
Entrepreneurship (low)	0.143	0.141	0.197	0.135
Entrepreneurship (mid)	0.137	0.136	0.187	0.141
Entrepreneurship (high)	0.161	0.164	0.181	0.154
Firm size (avg)	21.877	21.550	14.680	21.177

Counterfactual 3: Consumption Floor

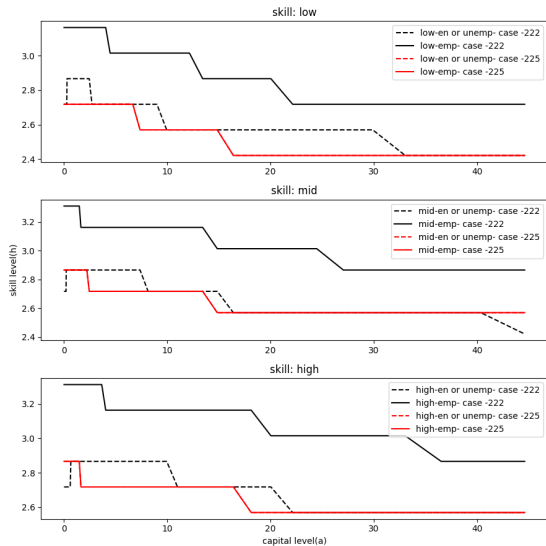
UI=0.4	Baseline	Lump-sum	Penalty
Entrepreneurship (avg)	0.145	0.144	0.144
Firm size (avg)	21.877	21.867	21.951
Unemp-Ent	0.00204	0.00189	0.00177

- ▶ At baseline UI, consumption-floor variants have modest effects on aggregate entrepreneurship.
- ▶ Effects are stronger when unemployment risk and insurance are jointly changed.

Cutoff Visualization



Baseline cutoffs



Baseline vs no-unemployment

Main Takeaways

- ▶ Necessity entrepreneurship is an equilibrium response to unemployment risk.
- ▶ UI reduces necessity entry and can improve allocation by discouraging low-productivity self-employment.
- ▶ Education heterogeneity matters for both entry and firm size responses.
- ▶ Policy design should account for both insurance benefits and occupational-sorting distortions.

Next Steps in Paper

- ▶ Finalize credit-friction wedge experiment.
- ▶ Finalize job-market weakening experiment (matching efficiency/productivity).
- ▶ Tighten baseline-data fit discussion for firm-size targets.
- ▶ Integrate final robustness checks into quantitative section.